End-to-End Data Management Pipeline for a Recommendation System

DML Assignment 2 | MovieLens 1M | Implementation and evaluation report

1. Objective and dataset

This project implements a reproducible movie recommendation pipeline covering data ingestion, automated validation, chronological preparation, feature generation, two recommendation models, ranking evaluation, experiment tracking and Prefect orchestration. The source dataset is MovieLens 1M from GroupLens. It contains user demographic fields, movie titles/genres and timestamped ratings. Age is retained as the dataset's age-group code, not interpreted as an exact age.

2. Pipeline design

The workflow downloads and normalizes the original MovieLens DAT files into three CSV entities: users, items and interactions. Validation checks missing values, duplicate entity/event keys, numeric types, rating range (1–5), timestamp validity and user/item referential integrity. Invalid rows are quarantined under data/errors and valid records continue. Preparation uses a global chronological 80/20 split; all derived summaries, features and models use training interactions only.

3. Data quality and preparation

Dataset	Checked	Valid	Quarantined
Users	6040	6040	0
Items	3883	3883	0
Interactions	1000214	1000207	7

The chronological cutoff is 2000-12-02T14:52:18+00:00. Training contains 800,168 rows (80.0%); test contains 200,039 rows. The actual ratio may differ slightly from 80/20 when timestamps tie.

4. Feature management

A reusable feature layer exports user features (activity count, average rating, preferred genre, recency and source demographics), item features (genre, interaction popularity, positive interaction count and average rating), and training interaction features (rating, positive-feedback flag, UTC weekday/hour and recency relative to the training cutoff). A feature catalog is saved in CSV and Markdown.

Model training, evaluation and orchestration

5. Recommendation models

The baseline ranks items by training positive-rating count, then total training interactions and average rating. The personalized model is item-item collaborative filtering: cosine similarity is computed between item vectors from a binary matrix of positive training ratings. Previously seen items are removed from each user's candidate list; popularity is used as a fallback/tie-breaker.

6. Evaluation results

Model	P@5	R@5	NDCG@5	Coverage	Diversity
Popularity baseline	0.2316	0.0314	0.2417	0.0246	0.7466
Item-item collaborative filtering	0.2483	0.0384	0.2624	0.1237	0.7250

Ranking metrics are macro-averaged across 1,122 users with at least one positive held-out rating for an item available in the training catalog. K=5; relevance is rating >= 4.0. Precision@K divides hits by K, Recall@K divides hits by the user's relevant held-out items, and NDCG@K discounts hits by rank. Coverage is the fraction of training catalog items recommended to at least one evaluated user. Diversity is mean intra-list Jaccard distance over genre sets.

6.1 Rating prediction evaluation (RMSE and MAE)

The personalized item-item model also estimates explicit 1–5 star ratings for held-out user–movie interactions. RMSE (root mean squared error) penalizes larger prediction errors more strongly, while MAE (mean absolute error) reports the average absolute difference between predicted and actual ratings. Lower values indicate more accurate rating predictions. These metrics are computed on held-out test ratings, not on ratings used to build the model.

Model	RMSE	MAE
Popularity baseline	N/A	N/A
Item-item collaborative filtering	1.0714	0.8207

N/A means that the model does not produce explicit user-specific rating predictions or that the corresponding metric was unavailable in the evaluation output.

7. Orchestration and reproducibility

Prefect orchestrates ingestion → validation → preparation → transformation → feature generation → model training → evaluation/reporting. Task retries are configured, and the two models are trained as separate tasks. MLflow records experiment parameters, metrics and artifacts using a local tracking store. The full workflow is launched with one command documented in README.md.

8. Limitations and considerations

55 positive held-out interactions involving items absent from the training catalog are excluded from ranking evaluation, and 55,969 positive held-out interactions belonging to users without training history are separately counted. Neither model can recommend an unseen catalog item or personalize for a user with no history. The collaborative filter is item-item and uses genre metadata only for diversity measurement, not model scoring. New users/items therefore remain cold-start cases. The MovieLens 1M data reflects historical ratings and should not be interpreted as current viewing preferences. Results depend on the chosen chronological cutoff, positive-rating threshold and K.

Conclusion

The implementation demonstrates the complete data management lifecycle and compares a global popularity baseline with a personalized collaborative-filtering recommender. The saved validation logs, feature catalog, model artifacts, evaluation table, MLflow runs and Prefect execution history provide traceable evidence for the pipeline.